Control Improvement Progress Report V2 (Adi)

sEMG implementation + EKF concept

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This is an internal status note, not a polished deliverable. It summarises where the "smarter tremor control" work stands: what's built, what's still open, and what tripped us up along the way. Kept light on the maths on purpose — the technical detail lives in `docs/CONTROL_IMPROVEMENT_PLAN.md`, `docs/SEMG_IMPLEMENTATION.md` and `docs/SEMG_DATASET.md` for anyone who wants it.

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1. The problem we're trying to fix

The spoon's current controller assumes it already knows the tremor frequency and filters out a fixed band (roughly 4–12 Hz). That works well when the assumption holds — about 93% tremor cancellation on average. But it breaks in two specific and important situations:

- When the person makes a deliberate hand movement (e.g. reaching for the plate) **at the same time** as tremor, the controller gets confused and cancellation drops to ~51%.
- When tremor sits outside the assumed 4–12 Hz band, or the hand moves suddenly (a step change), the controller actively makes things **worse than doing nothing** — in one test case, -197% (i.e. it amplifies the shake rather than cancelling it).

So while the headline number looks good, the honest average across all realistic test cases is closer to 64%, and it can fail badly in the exact moments (voluntary reach, sudden motion) that matter most in real use.

2. Why we're introducing an EKF (in plain terms)

An **EKF (Extended Kalman Filter)** is a mathematical technique for estimating something you can't measure directly — here, "is this movement tremor or is it the person's own intention?" — by continuously updating a best guess as new sensor data comes in, rather than relying on a fixed, pre-set filter.

Two things make it a better fit than the current fixed-band filter:

1. **It adapts.** Instead of assuming tremor is always 4–12 Hz, it estimates the actual tremor frequency in real time, so it can handle a wider range of tremor speeds and adjust automatically if it changes.
2. **It knows when it's unsure.** An EKF naturally produces a "confidence" signal alongside its estimate. When that confidence drops (e.g. during a sudden voluntary movement), we can

have the system fall back to "just follow the hand" rather than fight it — turning a dangerous negative-cancellation failure into a safe, neutral one.

We looked at simpler adaptive-filter alternatives used in tremor research (WFLC, BMFLC) but picked the EKF specifically for that self-confidence signal, since detecting "I'm currently wrong" is what the two failure cases above actually need.

3. What's done so far

EKF core, built and working standalone. The estimator itself — a 13-variable model that separately tracks "intended movement" and "tremor" from the handle's motion sensor — is implemented and passes its own correctness tests (frequency tracking, stability, repeatability).

A real bug found and fixed in that estimator. While investigating why the EKF struggled on the "voluntary movement + tremor together" test case, we traced it to a genuine coding defect: the estimator's internal "best guess" for hand position and its internal "uncertainty" about that guess were being updated on two different, inconsistent assumptions. Once corrected, that specific failure case improved from 30% cancellation to ~84–88% in testing. This is now an optional, switchable fix — not yet the default — pending a full side-by-side comparison against the current controller.

A second, independent sensor channel investigated: sEMG. sEMG (surface electromyography) measures the electrical signal produced by forearm muscles. The idea: muscle activity happens *before* the hand physically moves, so it could give the controller earlier warning of tremor and of intentional movement than motion sensors alone. We built:

- A simulated sEMG signal generator and a realistic "sensor front end" model (amplification, filtering, noise, etc., matching what a real sEMG sensor circuit would need to do).
- A full processing pipeline for extracting two useful features from that signal: overall muscle effort level, and rhythmic (tremor-frequency) muscle activity.
- On simulated data, this showed real promise: the muscle signal's tremor content matched the true tremor frequency closely, and — more importantly — showed the tremor "coming" in the muscle signal roughly 35 milliseconds before it showed up as physical shake. That head start is the entire reason sEMG is interesting.

Validated against a real, published sEMG dataset (not just simulation). We ran the sEMG pipeline against real human forearm muscle recordings (GRABMyo, a public research dataset from the University of Waterloo, via PhysioNet). Findings:

- Basic signal quality checks pass on real data.
- The specific question of whether two nearby muscle-sensor channels "bleed into" each other (cross-talk) — a key assumption behind how we planned to place the sensors — was tested directly. Result: it depends heavily on *where exactly* the two sensors are placed on the forearm. Placed on opposite sides of the arm, cross-talk is acceptable; placed close

together, it isn't. This is a genuinely useful, real finding that will directly shape the physical sensor placement design.

Tested whether sEMG actually improves the controller — result: not yet, and here's why. We ran a controlled experiment feeding the sEMG-derived tremor frequency into the EKF as an extra input. The result did **not** support the idea, for two clear reasons we could pin down precisely:

- Even with a *perfect*, error-free tremor frequency estimate, the "voluntary + tremor" failure case above only improved by about 8 percentage points. In other words, the tremor frequency estimate was never really the main bottleneck for that failure case — the coding bug described above was the real cause, and fixing that gave a much bigger improvement on its own.
- The sEMG-based frequency estimate itself turned out to be biased (systematically off) in a way that, when fed into the estimator, made several *already-working* cases noticeably worse.
- This was a useful experiment to run and rule out, but it's a genuine negative result, not a dead end for sEMG generally — see Section 5.

4. What's still pending

- **Decide whether to make the EKF the default controller.** It has not yet been run through the full 20-scenario benchmark that the current controller was validated against (only individual test cases so far). That full run, plus tuning, is the next concrete milestone.
- **The "confidence fallback" and "current/power budget" safety layers** described in the design plan (fall back to raw hand-following when uncertain; avoid demanding more motor current/voltage than the hardware can supply) are designed but not yet wired in and tested together with the EKF.
- **sEMG is validated as a signal source, but not yet fused into the control loop at all.** Everything sEMG-related so far runs alongside the controller for evaluation purposes only; none of it is currently used to influence the spoon's actual behaviour.
- **Three of the six real-data validation checks for sEMG remain unanswered**, because no existing public dataset combines forearm muscle signals with an actual physical tremor signal recorded at the same time (see Section 5 for why). Until that's addressed, several of the sEMG-related conclusions rest on simulation only, not real physiology.
- **The final side-by-side comparison report** (old controller vs. new, full results table, honest list of limitations) is planned but not started, pending the benchmark run above.

5. Problems / obstacles encountered

Finding a suitable dataset was harder than expected, and remains only partly solved. What we actually need is: real forearm muscle signals, recorded *at the same time* as a real physical tremor measurement (e.g. from a motion sensor), from a real tremor patient or a realistic

tremor simulation on a healthy person. We could not find any public dataset that provides all of that together:

- One promising Oxford dataset turned out to be an active clinical trial with no data released yet.
- A related, already-published Oxford dataset had only motion data, no muscle signal at all.
- The closest published research using exactly this kind of combined data exists, but the underlying data itself was never made public — it's private to that research group.
- We settled on GRABMyo (real muscle signals, real humans, publicly available, permissively licensed) as the best available substitute — but it only contains deliberate hand gestures, not tremor. That's why it could only validate 3 of our 6 planned checks; it simply isn't a tremor dataset, and we've been careful not to overstate what it proves.

Two real data-quality bugs found and fixed in the process:

- The simulated sensor "front end" was configured with settings that would have clipped (distorted) a real signal if it had been actual hardware — found by checking real signal statistics, and corrected.
- A subtler bug where an electrical filter started from an incorrect internal state caused every simulated sensor channel to briefly spike at the very start of each recording, potentially masking the clipping problem above. Also fixed.

A "promising-looking" cross-talk result earlier in the project turned out to be measured wrong, and we had to walk that back once it was checked properly against real data (see `SEMG_DATASET.md` for the full account). The corrected version is the one reported in Section 3 above. We'd rather report that correction transparently than leave the earlier, overly optimistic number standing.

The sEMG fusion idea, our leading hypothesis for how to fix the hardest failure case, did not pan out on the first attempt (Section 3, last item). This isn't necessarily the end of that idea, but it means the "muscle signal gives an early warning" concept needs a different implementation approach before it can help, not just plugging the same estimate into the same place.

6. Planned next steps

3. Run the corrected EKF (the version with the intent-tracking bug fixed) through the full 20-scenario benchmark used to validate the current controller, so we have a fair, complete comparison rather than a handful of hand-picked test cases.
4. Wire in and test the two safety layers together with the EKF: the "fall back to hand-following when uncertain" behavior, and the "don't demand more motor power than the hardware allows" behavior.
5. Tune the EKF's settings using that benchmark, then freeze them and re-run once more to confirm the numbers are reproducible.

6. Decide, based on that data, whether the EKF replaces the current controller, and write up the final before/after comparison document.
7. For sEMG: since the public dataset gap is the real blocker for the remaining validation checks, the plan is to record our own short synchronised session — muscle sensors plus a motion sensor, worn at the same time — once the physical sensor placement is finalised (informed directly by the real-data cross-talk finding in Section 3). This would let us test the two checks that specifically need a tremor-plus-motion-sensor combination, which no existing public dataset offers.
8. Revisit sEMG fusion with a different approach once the EKF itself is finalised and benchmarked — the first attempt's negative result was informative about *what doesn't work* (feeding a biased frequency estimate into the same slot), not proof that muscle signals can't help at all.